GG-B-00120b (DSA-DM)
April 16, 1969
Interim Revision of
Fed. Spec. GG-B-120a
September 20, 1956

INTERIM FEDERAL SPECIFICATION

BASEPLATE, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-B-120a.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers baseplates used for making rigid trial plates and individual impression trays in dental work.

* 1.2 Classification. Baseplates shall be furnished in the following two classes as specified (see 6.1).

Class 1. Upper form.
Class 2. Lower form.

* 2. APPLICABLE DOCUMENTS

* 2.1 Specifications, standards, and drawing. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6520

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SPECIFICATIONS

Federal

*	MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
*	PPP-B-566	Boxes, Folding, Paperboard.
	PPP-B-585	Boxes; Wood, Wirebound.
	PPP-B-601	Boxes; Wood, Cleated-Plywood.
	PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
*	PPP-B-636	Box, Fiberboard.
	PPP-B-676	Boxes; Set-Up, Paperboard.
*	PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (For Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking For Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

* (Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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ge shall be [illegible] interior packages, each unit and intermediate packa

STANDARDS

Military

* MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

20234 Baseplates, Dental

(Copies of Military specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The baseplates shall consist essentially of high grade shellac and mica. Composition shall be uniform throughout.

* 3.2 Style, design, and dimensions. Baseplates shall conform to Defense Medical Supply Center (DMSC) Drawing 20234 for style, design,and dimensions.

3.3 Color. Baseplates shall be pink in color.

3.4 Cooling. When tested as specified in 4.4.1, baseplates shall readily revert to their original shape.

3.5 Adaptability. Baseplates shall be easily adapted and shall not tear, blister, discolor, nor stick, when heated to a talc dusted test model; and shall neither warp nor lift away from the test model at any point, when tested in accordance with 4.4.2. Baseplates shall not char nor burn when luted to test model with a hot spatula.

3.6 Transverse strength. Transverse strength shall be no less than 1.75 kilograms (kg.) when tested as specified in 4.4.3.

3.7 Bending. The specimen shall bend 10 $\pm$ 5 millimeters (mm.) when tested as specified in 4.4.4.

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* 3.8 Spring back. The opening shall not exceed 65 mm. when tested as specified in 4.4.5.

* 3.9 Working qualities. Baseplates, when tested as specified in 4.4.6, shall comply with the following requirements:

(a) Baseplates shall be capable of being adapted to a wet or dry (without talcum dust) stone cast without adhering to cast.

(b) Heated baseplates shall not adhere to fingers when adapting to cast.

(c) The finished adapted baseplate (tray) shall not distort when immersed in water for 10 minutes.

* 3.10 Stability. The baseplates shall comply with the specification for at least 30 months from date of shipment, when stored between 50 and 80 degrees Fahrenheit.

3.11 Workmanship. Workmanship shall be first class throughout. Surfaces of the baseplates shall be smooth, and baseplates shall be free from blisters, cracks, or crinkles. The baseplates shall be free from defects which would detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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* 4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

* 4.2 Quality conformance inspection. The tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in table II (see 4.2.3).

* 4.2.1 Sampling for visual examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Inspection levels and Acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one box (unit of issue) of baseplates.

* Table I. Sampling for Visual Examination

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor	II	4.0

* 4.2.2 Sampling for examination of dimensions and quantity. Sampling for dimensional examination and for examination of quantity present in unit of issue (12 baseplates) shall be conducted in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level of 4.0 (percent defective).

* 4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with table II. For class 1, upper baseplates, no less than 3 baseplates for each test shall be taken for tests from each lot (control) number. For class 2, lower baseplates, no less than 3 baseplates for each test shall be formed into upper plates for test purposes from the same composition lot from which the lower baseplates are made.

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* Table II. Tests

Characteristic	Reqmt.	Test Procedure	Units of Product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Cooling	3.4	4.4.1	3	1 1/	3
Adaptability	3.5	4.4.2	3	1 1/	3
Transverse strength	3.6	4.4.3	3	3 2/	9
Bending	3.7	4.4.4	3	1 1/	3
Spring back	3.8	4.4.5	3	3 2/	9
Working qualities	3.9	4.4.6	3	1 1/	3

1/ A lot shall be rejected if any determination fails the test requirement.

2/ A lot shall be rejected if the average of the determinations per unit of product fails the test.

* 4.3 Examination. Baseplates shall be examined for defects including, but not limited to, those listed in table III.

* Table III. Classification of Defects

Categories	Defects
Major:	
101	Class not as specified.
102	Color not pink.
103	Cracks over 1/4 inch present in baseplates.
Minor:	
201	Surface not smooth and not free from blisters or crinkles.
202	Cracks under 1/4 inch present in baseplates.
203	Packaging and packing not as specified.
204	Marking not as specified.

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4.4 Tests.

* 4.4.1 Cooling test. Baseplates shall be flamed or heated in an oven, and then shall be cooled. After cooling, baseplates shall readily revert to their original hardness.

4.4.2 Adaptability. A test specimen of baseplate material shall be placed on a test model that has been preheated to 90 ± 2 degrees Centigrade (°C.) for at least 1 hour, and shall then be placed in an oven for 10 minutes at 90 ± 2° C. Remove test model and specimen and allow to cool for 1 hour to room temperature. The test model shall be constructed of maple wood to the dimensions as given in figure I.

4.4.3 Transverse strength. A test specimen of baseplate material shall be suspended across a 2-inch bridge using 3/16-inch glass rods for supports. Weight shall be applied in the middle of the strip by a plunger with its end consisting of an edge 3/32-inch radius and 1/2 inch long. Shot shall be poured into a can on top of the plunger at a rate of 1.5 kg. per minute until the specimen breaks. An average of three weights of the plunger and shot shall be taken as the transverse strength.

4.4.4 Bending. Place a whole upper plate on a smooth flat surface and flame with a Bunsen burner until softened; turn over and flame again until softened. Put aside to set at room temperature (21° to 27° C.) for exactly 1 hour. After 1 hour, place the baseplate in an oven at 37° C. ± 1/2° C. for exactly 10 minutes. The baseplate shall stand on edge in a narrow slot so that it will be heated evenly. When heated for 10 minutes, place the plate in a horizontal position with one slanting edge inserted into a slot 7/16 inch deep and just a little thicker than a baseplate to permit easy insertion and removal. Attach a weight and spring (figure II), totaling 34 to 35 grams, to the baseplate at the furthest point from the slot so that the edge of the weight enclosed by the spring shall be flush with the edge of the baseplate. The slot shall be above the floor of the oven a sufficient distance to allow for sagging of the baseplate when heated. The oven shall be regulated so that the temperature is constant at 37° C. when thermometer bulb is at the same level as the baseplate. The baseplate shall remain under the weight for exactly 5 minutes and shall then be removed to room temperature to cool until rigid. Measure with a scale the number of millimeters the edge of the plate has warped from the horizontal. This shall be done by placing the flat surface, which was in the slot, on a glass plate so that the warped section extends upward. Measure the height of the other edge from the glass.

* 4.4.5 Spring back test.

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* 4.4.5.1 Apparatus. The apparatus shall consist of two brass disks, one 5 inches in diameter and 1/4 inch thick, and the other 1-1/4 inch in diameter and 1/2 inch thick. The disks are tapped and joined together at the center of each by means of an 8-32 flat countersunk head screw extending through the large disk and approximately 3/4 of the way through the small disk.

* 4.4.5.2 Test method. Submerge the apparatus at least 2 inches below the water level of a water bath held at 55 ± 1° C., with the large disk facing down. Cut a flat strip specimen 1/2 inch by 3-1/2 inches from the test baseplate. Condition the strip in the bath for 10 minutes. Wrap the flat surface of the specimen around the circumference of the small disk and release the hold within 5 seconds. Allow one minute in bath for specimen to spring back. Then remove the specimen with care not to distort it, and let cool to room temperature for one hour. Measure the opening from the outside edges of the specimen strip with a millimeter ruler or caliper. Run test on 3 specimens and average their values.

* 4.4.6 Working qualities. Prepare an edentulous cast of artificial stone and allow cast to set at least one hour before performing tests. Select the applicable baseplate and center it on the dry (undusted) stone cast, and gently heat (without charring) the top of the baseplate with a Bunsen flame, so that the heat penetrates only half the thickness of the baseplate. Turn the baseplate over, place in the same position on cast. Evenly heat the baseplate again, until the baseplate begins to flow into the shape of the cast by virtue of its own weight. Moisten the fingers with water, and, by alternately applying heat and finger pressure to the baseplate, adapt the baseplate to the shape of the cast. The baseplate shall not adhere to the fingers during the heating and adapting procedures. When the adaption is complete, lift from cast and examine for adherence of baseplate to cast. This same procedure shall be used in examining for adherence of baseplate to a wet model by soaking stone cast (model) in water for 2 minutes prior to any adapting. Immediately submerge baseplate in water at a temperature of 50 ± 5° F., for 10 minutes. At the end of this time, remove baseplate from water and adapt again to original stone cast and examine for distortion.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One box containing twelve baseplates, as specified, constitutes one unit of issue.

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* 5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Class	Unit package	Intermediate package	Exterior container
1 or 2	1 unit	12 units	144 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

* 5.1.3.1 Wrap. Each plate shall be wrapped in tissue paper or other non-abrasive material.

* 5.1.3.2 Unit package. Twelve wrapped plates shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

* 5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250) or metal stitches is permitted.

* 5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

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* 5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load and shall be constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table III, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table II, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

* 5.3.1 Civil agencies. Interior and exterior containers shall be marked in accordance with FED-STD-123.

* 5.3.2 Military agencies.

* 5.3.3 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. Marking shall include the legend:

"STORE IN A COOL PLACE (50° - 80° F.)".

* 5.3.4 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the following legend shall be shown:

"STORE IN A COOL PLACE (50° - 80° F.)".

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6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Selection of applicable levels of packaging and packing(see 5.1 and 5.2).

* 6.2 Specification changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

Defense Supply Agency - DM

Review activities:

Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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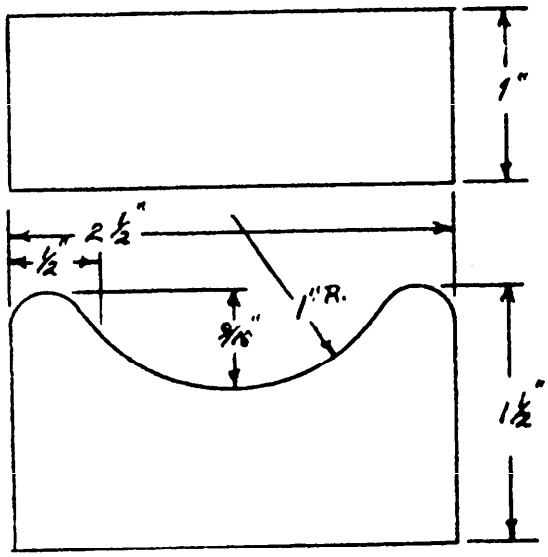


FIG. I

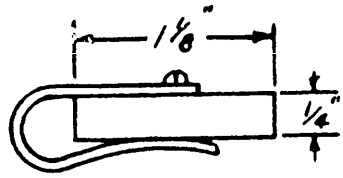


FIG. II

NOTE:
WIDTH OF CLIP ⅜"
MATERIAL - BRASS

TEST BLOCKS
DENTAL BASE PLATE

BASE PLATE, COMPOSITION, DENTAL.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

GG-B-00120b(DSA-DM) FSN

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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